CHEMISTRY

SECTION – I

Straight Objective Type

1. (c)

$$\frac{r_{He}}{r_{O_2}} = \frac{p_{He}}{p_{O_2}}\sqrt{\frac{M_{O_2}}{M_{He}}}$$

$$\frac{n_{He}/t_{He}}{n_{O_2}/t_{O_2}} = \frac{p_{He}}{p_{O_2}}\sqrt{\frac{M_{O_2}}{M_{He}}}$$

$$\frac{7.07 \times t_{O_2}}{1 \times 10} = \sqrt{\frac{32}{4}} = \sqrt{8} = 2\sqrt{2}$$

$$t_{O_2} = \frac{2 \times 14.14}{7.07} = \textbf{4 hr}$$

2. (b) Let the moles of $NaNO_3$, $NaNO_2$ and O_2 at equilibrium be a, b and c respectively and v_1 and v_2 will be the volume occupied by 'a' moles of $NaNO_3$ and 'b' moles of $NaNO_2$ respectively. They are small and can be ignored with respect to total volume i.e. (100 ml).

$$2NaNO_3(s) \rightleftharpoons 2NaNO_2(s) + O_2(g)$$

Moles at equilib.	a	b	c
Conc. at equilib.	$\frac{1000\,a}{v_1}$	$\frac{1000\,b}{v_2}$	$\frac{1000\,c}{100} = 10\,c = K_c$
Conc. after addn. of 30 g of $NaNO_2$ (v = 20 cc)	same	same	$\frac{1000\,c}{80} = \frac{100\,c}{8} = Q_c$, $Q_c > K_c$
Conc. after addn. of 30 g of $NaNO_3$ (v = 15 cc)	same	same	$\frac{1000\,c}{85} = Q_c'$, $Q_c' > K_c$

In both cases (addition of $NaNO_2$ as well as $NaNO_3$) the reaction will go in backward direction.

3. (b)

$$Ca(NO_3)_2(aq) + Na_2CO_3(aq) \longrightarrow CaCO_3(s) + 2NaNO_3(aq)$$

Initial millimoles	6	5	–	–
Millimoles after the reaction	1	0	5	10

$$CaCO_3(s) \rightleftharpoons Ca^{2+} + CO_3^{2-}$$

$$\frac{x+1}{100} \quad \frac{x}{100}$$

$$\approx \frac{1}{100}$$

$$Ksp = \left[Ca^{2+}\right]\left[CO_3^{2-}\right]$$

$5\times10^{-9} = \frac{1}{100}\times\left[CO_3^{2-}\right]$

$\left[CO_3^{2-}\right] = \mathbf{5\times10^{-7}M}$

4. **(a)** The reactions occuring during the electrolysis are

Cathode: $Sn^{2+} + 2e^- \longrightarrow Sn$

Anode: $Sn^{2+} \longrightarrow Sn^{4+} + 2e^-$

$2H^+ + 2e^- \longrightarrow H_2$

$$E_{H^+|H_2} = E^o_{H^+|H_2} - \frac{0.059}{2}\log\frac{(p_{H_2})}{[H^+]^2} = 0 - \frac{0.059}{2}\log\frac{1}{(10^{-7})^2}$$

$$= -0.059 \log (10^7) = -0.059 \times 7 = -0.413 \text{ V}$$

Moles of e^- used at cathode = Moles of e^- released at anode = $\frac{9.65\times30\times60}{96500} = 0.18$

Moles of Sn^{2+} oxidised = Moles of Sn^{2+} reduced = $\frac{0.18}{2} = 0.09$

Moles of Sn^{2+} left = 0.5 – 0.18 = 0.32

Concentration of Sn^{2+} left in solution = $\frac{0.32}{0.5}$ = **0.64 M**.

5. **(d)** The isomers are $[Cu\,(CN)_2\,(NCS)_2]^{2-}$

$[Cu\,(CN)\,(NC)\,(NCS)_2]^{2-}$

$[Cu\,(NC)_2\,(NCS)_2]^{2-}$

$[Cu\,(CN)_2\,(SCN)\,(NCS)]^{2-}$

$[Cu\,(CN)\,(NC)\,(SCN)\,(NCS)]^{2-}$

$[Cu\,(NC)_2\,(SCN)\,(NCS)]^{2-}$

$[Cu\,(CN)_2\,(SCN)_2]^{2-}$

$[Cu\,(CN)\,(NC)\,(SCN)_2]^{2-}$

$[Cu\,)(NC)_2\,(SCN)_2]^{2-}$

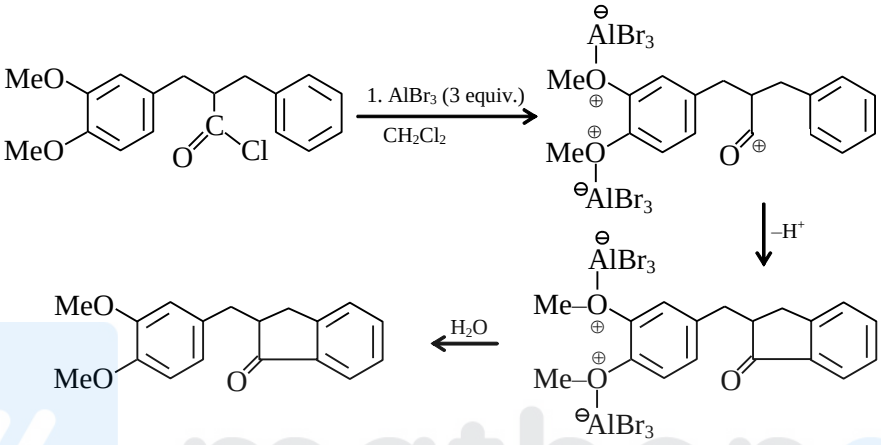
There are a total of **9** linkage isomers.

6. **(c)** O^{2-} is smaller than O_2^{2-}, thus, with a given metal ion (M^+), the lattice energy of M_2O would be greater than that M_2O_2. Peroxides would be preferably formed with larger cations because smaller cations will form a stable lattice of M_2O. Stability of metal chlorides, bromides and iodides increases down the group while that for fluorides decreases. The size of alkaline earth metal ions is smaller than that of alkali metal ions, so water of crystallization units, of alkaline earth metal salts are higher than that of alkali metal salts.

7. (b)

	NO_3^-	AsO_3^{3-}	CO_3^{2-}	ClO_3^-	SO_3^{2-}	BO_3^{3-}
$\frac{N}{2}$ value	3	4	3	4	4	3
No. of lone pair(s)	0	1	0	1	1	0
Shape	Trigonal planar	Trigonal pyramidal	Trigonal planar	Trigonal pyramidal	Trigonal pyramidal	Trigonal planar

8. (a)



One equivalent of $AlBr_3$ will be used to generate acylium ion and rest 2 equivalents react with MeO group to deactivate the ring (by forming Lewis acid-base complex having +ve charge on oxygen atom). Thus, acylium ion formed attacks on the other benzene ring (without MeO groups).

SECTION – II
Reasoning Type

9. (b)

$$SO_3^{2-} + \frac{1}{8}\overset{*}{S}_8 \longrightarrow \overset{*}{S}SO_3^{2-} \quad ; \quad {}^{-}O\!-\!\underset{\underset{\overset{\|}{S^*}}{\|}}{\overset{\overset{O}{\|}}{S}}\!-\!O^{-}$$

$$Ba^{2+} + \overset{*}{S}SO_3^{2-} \longrightarrow Ba\overset{*}{S}SO_3$$

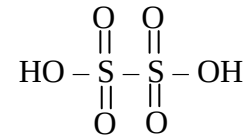
$$Ba\overset{*}{S}SO_3 + 2H_3O^+ \longrightarrow SO_2 + \frac{1}{8}\overset{*}{S}_8 + 3H_2O + Ba^{2+}$$

This reaction sequence indicate that the two sulphur atoms in thiosulphate are not equivalent and do not exchange with each other.

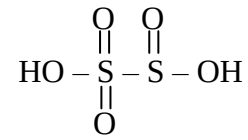
10. (b, c, d)

$H_2S_2O_4$: $HO - \underset{\|}{\overset{O}{S}} - \underset{\|}{\overset{O}{S}} - OH$

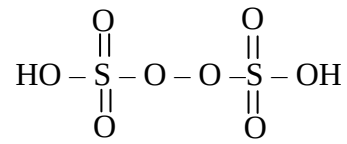
$H_2S_2O_6$:



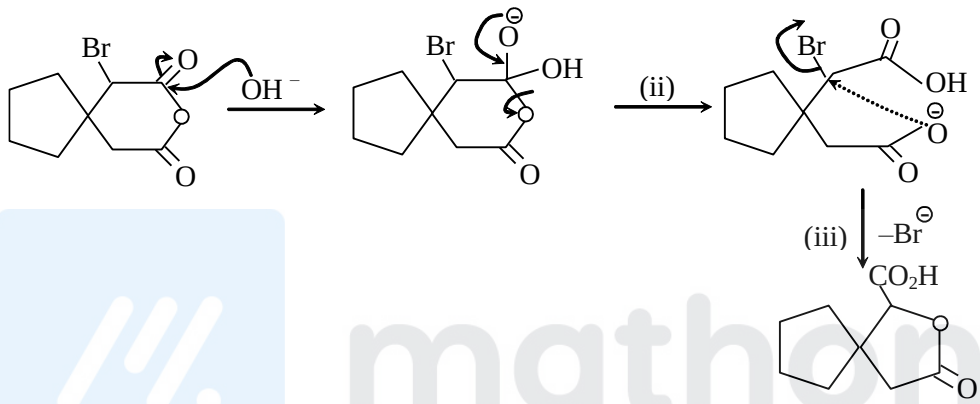
$H_2S_2O_5$:



$H_2S_2O_8$:



11. (a, b, d)



The first step of the reaction leads to nucleophilic substitution at acyl carbon while third step of the reaction involve nucleophilic substitution at alkyl carbon. Overall reaction leads to ring contraction from 6 to a 5-membered ring.

12. According to Dalton's law,

$p_{O_2} = P_{total} X_{O_2}^{gas} = 0.20 \times 760 = 152 \text{ mm Hg}$.

According to Henry's law,

$$p_{O_2} = k_H X_{O_2}^{liq}$$

$$152 = 3.04 \times 10^7 X_{O_2}^{liq}$$

$$X_{O_2} = 5 \times 10^{-6}$$

$$X_{O_2} = \frac{n_{O_2}}{n_{O_2} + n_{H_2O}} = \frac{n_{O_2}}{n_{O_2} + 1000/18} = \frac{n_{O_2}}{n_{O_2} + 55.56} \approx \frac{n_{O_2}}{55.56}$$

$\therefore$ Number of moles of O_2 dissolved $(n_{O_2}) = 55.56 \times 5 \times 10^{-6} = 2.77 \times 10^{-4}$

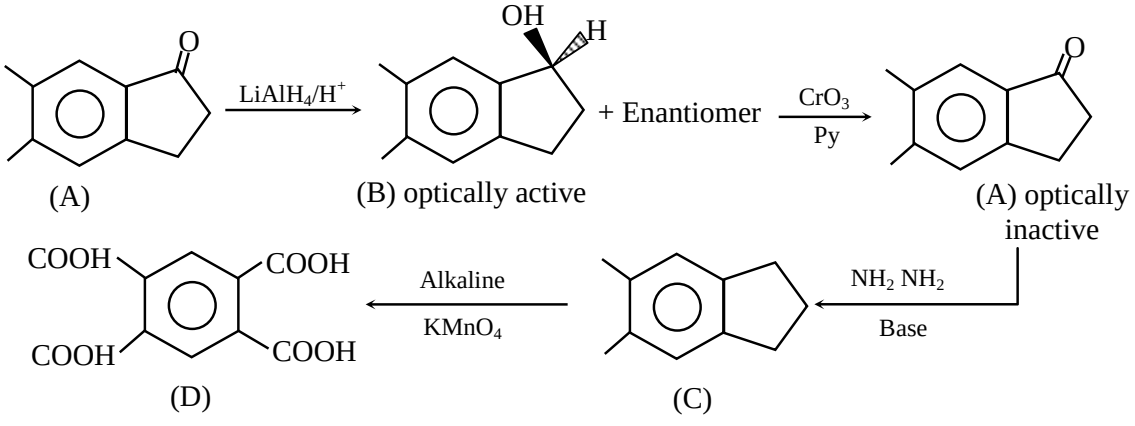
$[O_2] = 2.77 \times 10^{-4}$ M

$\therefore$ **(b) and (d)**

SECTION–III
Comprehension Type

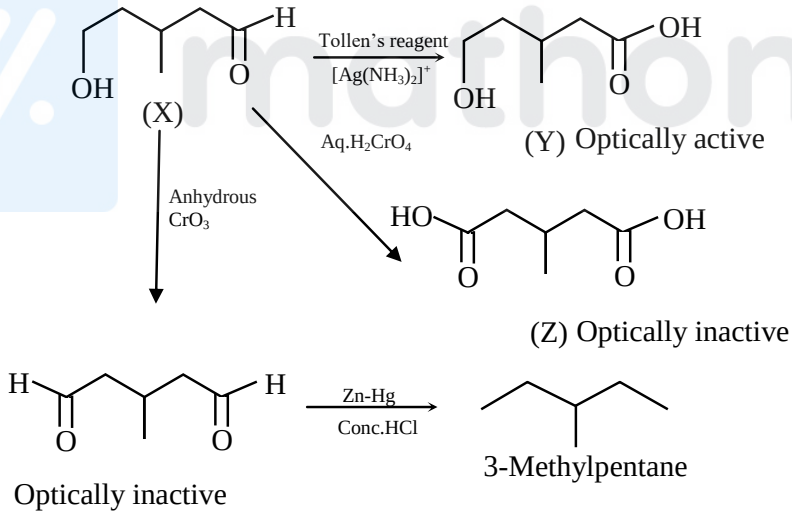
13. **(c)** Double bond equivalents of $C_{11}H_{12}O = 6$

No Tollen's test : Aldehyde absent, ketone may be present.

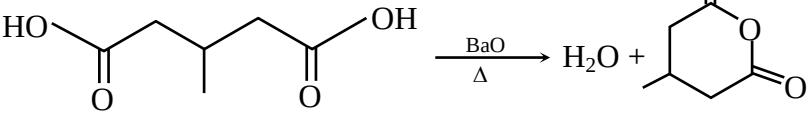


14. **(d)** $C_6H_{12}O_2(X)$ double bond equivalent= 1; Give positive Tollen's test.

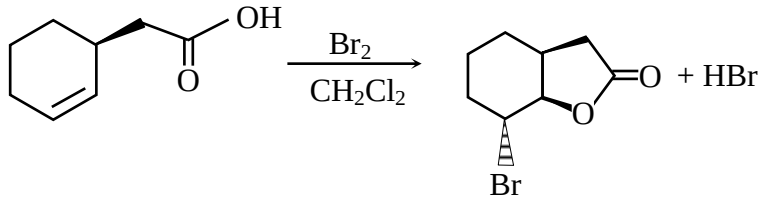
$\therefore$ Probable structure is

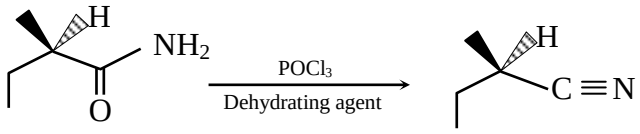


15. **(a)**

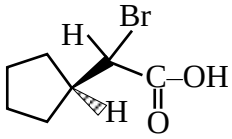


SECTION–IV
Matrix–Match Type

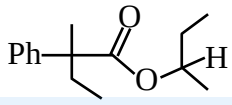
1. A - 

B - 

Optically active

C - 

racemic mixture

D - 

Diastereomeric mixture

Ans. (A): 2, 3, 4; (B): 3, 4; (C): 1; (D): 2, 3, 5

2. When phenolphthalein is used as indicator :

(A) Equivalent of HCl = Equivalent of NaOH + Equivalent of Na_2CO_3

$$= \frac{2}{40} + \frac{5.3}{106} = \frac{2}{20}$$

$\therefore$ Mass of NaCl $= \frac{2}{20} \times 58.5 = 5.85$

$\therefore$ No. of equivalent of HCl = $N_1\ V_1 \times 10^{-3} = \frac{2}{20}$

$\therefore$ $V_{HCl} = \frac{2}{200 \times 0.1 \times 10^{-3}} = \frac{2 \times 10 \times 1000}{200} = 100\,mL$

(B) No. of equivalent of HCl = $N_1\ V_1 \times 10^{-3} = \frac{1}{40}$

$\therefore$ $V_{HCl} = \frac{1}{40} \times \frac{1}{0.1} \times \frac{1}{10^{-3}} = \frac{10 \times 1000}{40} = 250\ mL$

(C) No. of equivalent of HCl = $\frac{5.3}{106} = \frac{1}{20}$

$\therefore$ $V_{HCl} = \frac{1}{20} \times \frac{1}{0.1} \times \frac{1}{10^{-3}} = \frac{10 \times 1000}{20} = 500\,mL$

(D) No. of equivalent of HCl = $\frac{1.06}{106} = \frac{1}{100}$

$\therefore$ $V_{HCl} = \frac{1}{100} \times 10 \times 1000 = 100\,mL$

When methyl orange is used as indicator:

(A) No. of equivalent of HCl = $= \frac{2}{40} + \frac{5.3}{53} = \frac{1}{20} + \frac{1}{10} = \frac{3}{20}$

$\therefore \quad V_{HCl} = \frac{3}{20} \times 10 \times 1000 = 1500 \text{ mL}$

(B) No. of equivalent of HCl = $\frac{1}{40} + \frac{2.1}{84} = \frac{2}{40} = \frac{1}{20}$

$\therefore \quad V_{HCl} = \frac{1}{20} \times 10 \times 1000 = 500 \text{ mL}$

(C) No. of equivalent of HCl = $\frac{5.3}{106} + \frac{2.1}{84} = \frac{5}{40}$

$\therefore \quad V_{HCl} = \frac{5}{40} \times 10 \times 1000 = 1250 \text{ mL}$

(D) No. of equivalent of HCl = $\frac{1.06}{53} = \frac{1}{50}$;

$\therefore$ Mass of NaCl = $\frac{1}{50} \times 58.5 = 1.17 \text{ g}$

$\therefore \quad V_{HCl} = \frac{1}{50} \times 10 \times 1000 = 200 \text{ mL}$

Mass of $CaCO_3$ formed :

$Na_2CO_3 + CaCl_2 \rightarrow CaCO_3 + 2NaCl$

$\therefore$ Moles of Na_2CO_3 = Moles of $CaCO_3$

$\therefore$ In (A), moles of $CaCO_3$ formed = $\frac{5.3}{106} = \frac{1}{20}$

$\therefore \quad W_{CaCO_3} = \frac{100}{20} = 5 \text{ g}$

In (C), Mole of $CaCO_3$ formed = $\frac{100}{20}$

$\therefore \quad W_{CaCO_3} = \frac{100}{20} = 5\text{g}$

In (D), $W_{CaCO_3} = \frac{1.06}{106} \times 100 = 1\text{g}$

Ans. **(A): 3, 5; (B) : 2; (C) : 1, 5 ; (D) : 4**

SECTION–V
Integer Answer Type

1.

$$CH_3NH_2 + HCl \longrightarrow CH_3\overset{\oplus}{N}H_3\,Cl^-$$

At equivalence point – – 40×0.15 = 6 mmole

$CH_3\overset{\oplus}{N}H_3$ undergo hydrolysis while Cl^- remain unhydrolysed.

$$CH_3NH_3^+ + H_2O \rightleftharpoons CH_3NH_2 + H_3O^+$$

$$\frac{6}{80}(1-h) \qquad \frac{6}{80}h \qquad \frac{6}{80}h$$

$$K_a = \frac{[CH_3NH_2]\,[H_3O^+]}{[CH_3NH_3^+]} = \frac{\frac{6}{80}h^2}{(1-h)} \simeq \frac{6}{80}h^2$$

$$h = \sqrt{\frac{K_a \times 80}{6}}$$

$$[H_3O^+] = \frac{6}{80}h = \sqrt{K_a \times \frac{6}{80}} = \sqrt{\frac{4}{3}\times 10^{-11} \times \frac{6}{80}} = \sqrt{10^{-12}}$$

$$[H_3O^+] = 10^{-6}\,M$$

$$\therefore\ pH = 6.0$$

2. To prepare a silicone containing 'n' Si–O–Si linkages, the total number of units required are (n+1), of which 2 are chain blocking units, R_3SiOH and (n–1) are chain building units, $R_2Si(OH)_2$. Thus, the number of $R_2Si(OH)_2$ units needed are **9.**

3. $R–OH + CH_3COCl \longrightarrow R–OCOCH_3 + HCl$

When an alcohol containing one OH group reacts with 1 mole of CH_3COCl, the molecular mass of the product will be 42 units more than that of ROH. If the original compound contains 'n' OH groups, then the molar mass of product will be 42n units more than that of original compound. In the present case,

166 + 42n = 334

$\therefore$ **n = 4**

Thus, the compound contains 4 – OH groups.

4. $\because$ dv = 1.0 pm^3

$\therefore\ P \propto \psi^2\,dv$, where $\psi^2 \propto e^{-2r/a_0}$

At nucleus, r = 0.

$\therefore\ P_{r=0} \propto e^0 . (1\ pm)^3 = 1\ (pm)^3$

At a distance $r = a_0$,

$$P_{r=a_0} \propto e^{-2}.(1.0\ pm^3) = 0.14 \times 1 pm^3$$

$$\therefore\ \frac{P_{r=0}}{P_{r=a_0}} = \frac{1}{0.14} = 7.1 = \mathbf{7}$$